

	Document Type:	Document Code	RES- PROP-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	November 25, 2019
	RESEARCH PROPOSAL	Page	Page 1 of 8

(1) Research Program/Project/Study Title: COMPUTER COMPETENCY OF FRESHMEN BSIT STUDENTS OF CAPIZ STATE UNIVERSITY-PILAR SATELLITE COLLEGE	
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Academic Rank:	INSTRUCTOR
Co-Authors (with academic rank):	Instr. PHILOMEL INNOCENT OBLIGAR Engr. SHARON B. BATICADOS
Campus/Satellite College/Address:	CAPIZ STATE UNIVERSITY - PILAR SATELLITE COLLEGE
Telephone/Cell Phone/Fax Number:	09778179979
E-mail Address:	bsit-pilar@capsu.edu.ph

Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education/Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research ☐ Developmental/Computer/Engineering/Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		PILAR
Province:		CAPIZ
Region:		VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 2 of 8

		☐ Collaborative Research (external agency)
(7) Sector/Commodity Students		(8) Discipline/Field of Specialization Computer
(9) Rationale Information and Communication Technology (ICT) has taken over the global economy. Most things if not all require ICT to be performed. The education sector is not left out of this as Information and Communication Technology (ICT) has improved the method of teaching and learning; it has made teaching and learning less difficult. Teaching and learning with the aid of ICT has grown beyond the four walls of a classroom to learning from the comfort of one's apartment. This advancement in technology has equally bridged the digital divide between information rich and poor society. Despite the fact that students are sometimes referred to as "digital natives," certain household traditions and socioeconomic variables can influence their capability to develop age-appropriate computer literacy. It is generally accepted that one of the keys to being successful is to be computer literate or to have a certain level of technical competence. Children are taught to use computers at a young age and as they proceed through primary and secondary grades they regularly come into contact with various technologies. Because of this, they become adept at being users of technology. Recently, there has been much research trying to identify basic computer literacy skills that are needed for all college-level students and to determine if a need exists for a general computer literacy/skills course for all undergraduate students. Incoming freshmen students frequently complain about having to take a literacy course because they feel like they already "know how" to use computers. Clearly, their view of what skills are necessary to be literate differ from what college educators believe. If, as is commonly believed, it is determined that these incoming students lack important skills and knowledge then the next question is how are these skills and concepts best taught? Should colleges and universities require that a formal course be taken in this area as part of their core curriculum? If so, what would such a course cover and how would it be assessed.		
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.) This study will be conducted to students enrolled this SY 2022-23. The team will conduct a pre evaluation skill test to the respondents before and after the intervention. MS assessment tool will be used to test their computer competency skills.		
(10) Objectives/Statement of the Problem Generally, the study aims to provide intervention to enhance BSIT students' computer skills. Specifically, this aims to find out the following: 1. Find out the socio-demographic profile of the respondents as to sex, age, parents educational attainment, parents occupation and parents monthly income. 2. Determine the respondents' STRANDS. 3. Determine the level of computer literacy of the respondents. 4. Determine the number of students incompetent to computer. 5. Determine the Significsnt difference on the computer literacy of the respondents when group according to socio-demographic profile. 6. Determine if there is significant relationship on the STRANDS and computer literacy of the respondents.		
(11) Scientific Basis/Theoretical Framework ICT and the FilipinoTeacher		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 3 of 8

The conditions of ICT utilization in education has been studied by UNESCO (2005). An examination of countries in the Asia-Pacific region has shown that ICT is not being used to its full potential in enhancing the quality of teaching and learning. There are technical and capacity-related barriers that need to be overcome. Many countries of the region do not make use of ICT at all in their education system due to technical barriers such as lack of infrastructure, equipment and internet connectivity. ICT is used simply as a supplement for existing pedagogical practices. In order to fulfill the potential of ICT as a tool for enhancing teaching and learning, ICT must be fully integrated into pedagogical processes, which requires a cognitive shift on the part of policy makers, educators, curriculum developers, and administrators (Amirin, 2007). This cognitive shift shall be discussed later in the section on frameworks of ICT in education.

In the Philippines, the government is seeking ways to maximize the full potential of information and communication technology (ICT) to boost the growth of the economy (Tana, 2004). The Philippines is recognized for its competitive advantage in the IT services sector, being the second largest producer of computer services in Asia, next to India (DTI, 2008). In line with this, the NEDA (2011) published the Philippine Development Plan. Some of the notable targets for ICT for the year 2016 are as follows:

- Increased percentage of public schools with broadband connection:
 - High Schools from 59.37 percent in 2010 to 100 percent in 2016
 - Elementary Schools from 0 percent in 2010 to 22.23 percent in 2016.
 - Increased percentage of households with broadband connection from 13.8 percent in 2008 to 25.8 percent in 2016.
- Increased Cellular Mobile Telephone System (CMTS) coverage as a percentage of:
 - Municipalities, from 94.7 percent in 2009 to 100 percent in 2016.
 - Barangays, from 85 percent in 2009 to 100 percent in 2016
- Increased average broadband speed, Mbps
 - Upload
 - Download
- Increased number of broadband subscriptions per 100 participants:
 - Fixed, from 1.87 in 2009 to 4.69 in 2016
 - Mobile, from 10.2 in 2009 to 28.24 in 2016
- Increased percentage of Local Government Units (LGUs) with broadband connection:
 - Municipalities
 - Barangays

Presently, the Philippine ICT industry is led by Information and Communications Technology Office of the Department of Science and Technology (DOST-ICTO) (Gualin, 2013). "The DOST-ICTO is the government arm tasked to lead government's efforts toward the rapid development and improved global competitiveness of the country's information and communications technology industry." The Philippines ICT service advantage needs a continuous pool of manpower... the DTI (2008) recognizes the need to incorporate IT in the basic levels of education by providing computers to public schools. Gualin (2013) reported that "The IT-BPM industry is one of the major economic drivers of the Philippines and is the country's fastest-growing industry. In 2012, it generated US\$13.2 billion revenues, posting a 19-percent increase from 2011, and created 137,066 net new jobs, which brings total employment in the industry to 776,794 at the end of 2012."

One of the more urgent tasks of our Philippine educational system is to fit our educational offerings to the demands of this IT-BPM industry and the blossoming careers that are ICT related. However, only a "few teachers are confident in using a wide range of ICT resources, and limited confidence affects the way the lesson is conducted. Many teachers still fear some forms of technology, which prevents them [teachers in] making much use of them in their teaching" (Cox and others, 2003). "Public school teachers provided with computer laptops and basic computer skills training are using it for administrative clerical tasks, but not to enhance their teaching in the classrooms" (United States Agency for International Development, USAID as reported by Ronda, 2011). USAID's study partnered with DepEd and pointed out that "Teachers newly equipped with basic computer skills and trained in information and communication technology (ICT) are using their new skills mostly for administrative tasks such as preparing DepEd-required reports, calculating grades and typing lesson plans. Only a few teachers use their newly-acquired skills and the computers in the classroom to actually improve the learning experience of the students in the core subjects of English, Math and Science" (Ronda, 2011).

(12) Review of Literature/Prior Art Search

Talandron-Felipe, M.M. (2020) focused on two key causes of the digital divide: socioeconomic status and location (urban vs. rural) (income clusters). The findings provided more proof that students' digital disparity is influenced by their geographic location and status. Both factors affect the possession of devices, the reliability of the internet connection at home, and the frequency of access. Previous online learning experiences also demonstrate the importance of a student's location. The relationship between location groups and alternate

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 4 of 8

methods of accessing the internet is important, with students from urban areas more likely to pay for a temporary data subscription, visit internet cafes, or utilize their neighbors', friends', and family members' Wi-Fi. On the other side, people from rural areas are more inclined to use free data and public Wi-Fi because they don't have access to any other forms of connectivity.

Wallace, P. et. Al. (2005) mentioned that many schools and universities require prospective students to demonstrate a specific degree of computer competency, according to Al. (2005). Many academic institutions now provide introductory computer courses to help students fulfill this requirement. Other schools are thinking about getting rid of their introductory computer classes, expecting students to pass a competence test and demonstrate adequate computer knowledge. This is based on the supposition that students have already acquired both computer ideas and applications in high school or through other personal experiences. Students can enroll in alternative compulsory, elective, or optional courses that will help them fulfill their degree requirements if they have the option to take a computer competence exam in place of a necessary course. However, students who might normally profit from the knowledge gained from such a course may struggle if the introductory computer skills course is not offered. In order to find out whether students have sufficient computer knowledge and skills to skip the introductory computer fundamentals course, a study of 140 incoming business freshmen was created. Students were assessed using online testing software on both Excel proficiency and computer principles that they should have mastered by the end of an introductory computer fundamentals course. The results showed that, on average, the pupils had test scores of 60%, which is far below the passing mark. Only 36% of the students who participated in this study's test were able to "test out" of the course, or exempt themselves from it. Following course instruction, students who were unable to receive an exemption from the course showed a mean gain or improvement of 26 points on the Excel post test and 23 points on the Computer Concepts post test. According to the data, course teaching was effective in achieving the targeted result of giving students who couldn't get an exemption relevant knowledge of information technology concepts and abilities. These results lead to the conclusion that new business students lack the computer expertise, talents, and information required to pursue their undergraduate degree programs. First-year business students are advised to take an introductory computer course that covers both information systems (IS) theory and information technology (IT) software applications.

Higntte, M. et al (2009) argued that information literacy is a concept that has evolved as a result of recent efforts to bring technology-based teaching and research to a level above the perennial concepts previously associated with "computer literacy"; the focus of information literacy training is on developing the skills of students to construct/collect and analyze information in a way that is the basis for effective decision making. Result of this study on 600 college students that students' information literacy achievement levels were modest at best-only 40% of the students' achieved a "passing" grade on the exam. Additional result shows significant differences in achievement among the various groups of students with, for example females scoring significantly higher than males on the exam.

According to Gupta, G. (2006), undergraduates work in today's computer-centric, information age core curriculum. The number of computers in use worldwide is increasing, especially in developing countries. Computers affect every aspect of our lives and every sector of global society. Employers prefer computer-savvy employees because they are more productive and efficient at work than non-computer-savvy employees.

Fisher (2000) stated that an ongoing challenge in teacher education is keeping courses up to date. This article outlines the current skills, attitudes, and perceptions regarding computer technology in the classroom from a preservice candidate perspective. The purpose of this study is to foster discussion on how to further develop pre-educational technology courses that meet the needs of an ever-evolving and changing student and teacher population in terms of optimizing computer technology skills. The primary instrument was a self-assessment survey of skills, attitudes, and perceptions regarding the use of computer technology in the classroom among new students entering introductory courses at major Midwestern United States colleges during the fall 1999 semester. A total of 205 students were surveyed. To increase participation, questionnaires were distributed to trainee teachers at the first course rather than mailed to their homes. Respondents were categorized as either 'Beginner' or 'Experienced Users' of computer technology based on their responses to the skills section of the survey. This article provides an overview of the results and categorizes them according to three general issues: general attitudes toward teaching using computer technology, intended use of computers when beginning to teach, and perceived problems to be expected with computer technology when beginning to teach. Finally, an informal discussion of student expectations for future teacher education programs and the implications for school districts are presented from the free-response section of the questionnaire.

(13) Methodology

Research Design

This research study will use the descriptive-correlational research design. This aims to describe the level of students' computer competency using MS Office Assessment Tool.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 5 of 8

Locale of the Study

This study will be conducted at Capiz State University Pilar Satellite College. Capiz State University offers 6 Bachelors courses namely: Bachelor of Science in Business Administration, Bachelor of Science in Social Work, Bachelor of Science in Agriculture, Bachelor of Elementary Education, Bachelor of Science in Hospitality Management and Bachelor of Science in Information Technology

Respondents of the Study

The respondents of the study will be the first year BSIT students of Capiz State University enrolled during the SY 2022-2023.

Sample Size and Sampling Technique

The respondents of the study will be the first year BSIT students enrolled during the SY 2022-2023. Thirty (30) students will be selected from the target population using simple random sampling technique,

Research Instruments

The instruments that will be used in gathering data will be the questionnaire and the MS Office Assessment Tool. The questionnaire will elicit information on the socio demographic profile and the senior high school STRAND of the respondents and the MS Office Assessment Tool will be used to evaluate the level of computer competency of the respondents. A skills test on basic computer hardware operation will be conducted.

Data Gathering Procedure

Before the actual gathering of data, a letter of permission from the office of the Campus Satellite Director will be obtained. As soon as permission will be granted, the questionnaire will be distributed to the respondents at the same time the MS Office literacy assessment will be administered to them. Data gathered will then be tabulated and analyzed using statistical tools.

Statistical Tools

Statistical tools such as mean and percentage will be used to analyze the data on the respondents' socio demographic profile, T-test for significant relationship, and ANOVA for significant difference.

(15) Expected Output

Data on computer literacy as basis for making Innovative Learning Materials

(15) Technology Transfer Activity Design

Innovative Learning Materials that will serve as learning guide for the instructional extension to incoming first year BSIT students.

(16) Target Beneficiaries

1. Incoming BSIT students
2. Senior High School Students of the feeder school

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Prof. KAIZELL MICKHOS A. GERSANIVA	Study Leader	Computer Science
Engr. SHARON B. BATCADOS		Computer Engineering
Instr. PHILOMEL INNOCENT OBLIGAR		Biological Science

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

((18) Budgetary Requirement

P20, 000.00

	Document Type:	Document Code	RES- PROP-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	November 25, 2019
	RESEARCH PROPOSAL	Page	Page 6 of 8

(19) Literature Cited(include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

- Talandron-Felipe, M.M. (2020). The digital divide among students and support initiatives in the time of Covid-19
- Gupta, G. K. (2006). Computer literacy: Essential in today's computer-centric world. ACM SIGCSE Bulletin, 38(2), 115-119.
- Higntte, M., Margavio, T. M., & Margavio, G. W. (2009). Information literacy assessment: Moving beyond computer literacy. College Student Journal, 43(3).
- Mercedes Fisher (2000) Computer skills of initial teacher education students, Journal of Information Technology for Teacher Education, 9:1, 109-123, DOI: 10.1080/14759390000200075
- Wallace, P. & Clariana, R.B. (2005). Perception versus Reality—Determining Business Students' Computer Literacy Skills and Need for Instruction in Information Concepts and Technology. Journal of Information Technology Education: Research, 4(1), 141-151. Informing Science Institute. Retrieved October 31, 2022 from <https://www.learntechlib.org/p/111567/>.

(20) Capsule Curriculum Vitae

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 25, 2019
			Page	Page 7 of 8



KAIZELL MICKHOS A. GERSANIVA

PROFILE

<https://kaizellmickhos.github.io/profile/>

PERSONAL SKILLS

- ✓ Hardworking, dependable and has the desire to help others.
- ✓ Willing to take on new challenges
- ✓ Open minded and able to work independently

PERSONAL INFORMATION

DATE OF BIRTH:
March 13, 1992

PLACE OF BIRTH:
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+639778179979

ADDRESS:
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Capiz 5800

EMAIL:
kaizellmickhoadam@gmail.com

CIVIL STATUS:
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LANGUAGE:
Hiligaynon, Tagalog, English

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE

Iloilo Science and Technology University
Lapaz, Iloilo City

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

S.Y 2008 - 2012
Filamer Christian University
Roxas Avenue, Roxas City, Capiz

WORK EXPERIENCE

CAPIZ STATE UNIVERSITY – PILAR SATELLITE COLLEGE

Position: Instructor I
Address: Natividad, Pilar, Capiz
Date: January 3, 2022 to Present

CAPIZ STATE UNIVERSITY – ROXAS CITY MAIN CAMPUS

Position: Emergency Instructor
Address: Fuentes Drive, Roxas City
Date: June 08, 2015 – December 22, 2021

SPRING VALLEY TECH CORP.

Position: Back Office Specialist
Address: Pueblo de Panay Township, Roxas City, Capiz
Date: May 02, 2019 – November 10, 2019

FILAMER CHRISTIAN UNIVERSITY

Position: Part-time Faculty
Address: Roxas Avenue, Roxas City, Capiz
Date: December 16, 2014 – March 30, 2015

SUYEN CORPORATION (BENCH)

Position: Store Supervisor, Unit Head
Address: Paliparan, Dasmariñas Cavite 4114
Date: July 07, 2012 – June 07, 2015

AWARDS AND ACHIEVEMENT

MICROSOFT EDUCATION AMBASSADOR

Microsoft Philippines

Computer Hardware Servicing NC II

Trainers Methodology Certificate I

	Document Type: DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
		Revision No.	00
	Document Title: RESEARCH PROPOSAL	Effective Date	November 25, 2019
		Page	Page 8 of 8

SEMINARS AND TRAININGS

Webinar: Understanding Autism

UP Visayas, May 18, 2022

Training Workshop on Patent Search for CAPSU Faculty Researchers

CAPSU (IPMO), May 17, 2022

Webinar In Basic Motion Graphics using Adobe After Effects

DICT Region 9, May 11, 2022

Introduction to Python Programming

Negros Occidental Language and Information Technology Center (NOLITC), April 21, 2022

Seminar on Making Instructional Material in New Normal

CAPSU – Pilar, March 31, 2022

RDE 5-year Agenda Setting and Roadmap Development for Research, Extension, and IPMO

CAPSU (Office of the VP for RDE), March 21-22, 2022

Training on Intellectual Property Basics for CAPSU Faculty Researchers and Students

CAPSU (IPMO), February 24, 2022

Psychosocial Support Training for Teachers and Other Mental Health Advocates

CAPSU – Pilar (Guidance Office), January 05, 2022


Katzell Mickhos A. Gersaniva